David Bedolla, PhD

Robotics Engineer | Manipulation, Control & Robot Learning

Portfolio | LinkedIn | GitHub | davidbedollamartinez@outlook.es | Montréal, QC, Canada

PROFESSIONAL PROFILE

Robotics Engineer specializing in robot manipulation, control, and robot learning, with a strong foundation in kinematics, dynamics, real-time systems, and physical robot deployment. Develops ROS 2/C++ whole-body and Cartesian control for a physical 10-DoF mobile manipulator and previously deployed learning-based control, inverse kinematics, and 1-4 kHz motion control on a 7-DoF rehabilitation exoskeleton. Connects mathematical models and simulation to hardware validation, sensing, safety, and system debugging.

PROFESSIONAL EXPERIENCE

Associate Researcher

Mar 2026-Present | Lab INIT Robots, Montréal

- Develop and iterate a 400 Hz ROS 2/C++ whole-body motion-control pipeline for a physical 10-DoF AgileX and Kinova Gen3 mobile manipulator.
- Implement analytical and Pinocchio Jacobian controllers for Cartesian response, elbow posture, joint limits, filtered odometry, arm-base coordination, and repeatable trajectory execution.
- Integrate command validation, safe state transitions, world-frame tracking, PS4 teleoperation, simulator-based tests in Gazebo and MuJoCo, and a common ROS 2 tactile-map interface adapted from Lab-CORO TSF-85.

Postdoctoral Researcher

Mar 2025-Mar 2026 | École de technologie supérieure (ÉTS), Montréal

- Diagnosed a real-time control bottleneck and reduced lower-limb robot latency by 30 times, enabling torque control on target hardware. Improved ROS/Python scheduling for vehicle testing.
- Developed symbolic dynamics and inverse-kinematics tools and delivered 6-DoF IK and motion-control support to international assistive robotics teams.

Research Professor

Mar 2024-Mar 2025 | Universidad Tecnológica de la Mixteca, Mexico

- Developed and tested 7-DoF controller optimization that reduced tracking error by approximately 20 percent and contributed supervision to adaptive-gripper research.

Research Assistant

Jan 2020-Dec 2023 | École de technologie supérieure (ÉTS), Montréal

- Deployed and tuned a 1-4 kHz LabVIEW RT/FPGA and C++ stack on a physical 7-DoF exoskeleton, including BLDC actuators, motor drivers, DAQ, safety logic, and trajectory execution.
- Implemented 50 microsecond FPGA PI current regulation, Hall-feedback position tracking, and torque-to-current commands. Characterized friction, deadzones, backlash, and tracking response on hardware.
- Integrated ATI Nano17 force/torque sensing, admittance control, motor-current-based external-force observation, and IMU, EMG, and depth-camera motion acquisition.
- Deployed Gaussian Process residual modeling in robust MPC at approximately 387 microseconds for a 10-step horizon and used predictive uncertainty to adapt robustness margins.
- Achieved approximately 5 microseconds for analytical inverse kinematics, 43 microseconds for learned inverse kinematics, and 210 microseconds for feedforward torque. Reduced symbolic expressions by approximately 90 percent.

Research Professor

Oct 2018-Oct 2019 | UNSIJ, Oaxaca, Mexico

- Implemented real-time neural-network inverse kinematics for a 6-DoF PUMA robot and MATLAB CNN experiments for wildlife detection in camera-trap datasets.

Professor

Feb 2017-Oct 2018 | TecNM, Puebla, Mexico

- Deployed a TI-controller hardware-in-the-loop manipulator emulator at a 10 microsecond sampling period and implemented FPGA voice processing plus computer-vision experiments.

FLAGSHIP ROBOTICS PROJECTS

Whole-Body Mobile Manipulation

Problem: hold a world-frame end-effector target while the base moves. **Contribution:** odometry transforms, arm-base coordination, redundancy handling, and teleoperation. **Hardware:** physical AgileX/Kinova. **Stack:** ROS 2, C++, Pinocchio. **Result:** onboard 400 Hz deployment. Video.

Kinova Gen3 Cartesian Control and Redundancy

Problem: coordinate Cartesian motion and elbow posture on a redundant physical manipulator. **Contribution:** combined analytical redundancy with Pinocchio Jacobian velocities, joint-limit handling, command arbitration, and PS4 teleoperation. **Result:** approximately 3 microseconds for the analytical redundancy calculation. Video.

ETS-MARSE Learning-Based Control and Human-Centered Motion

Problem: constrained exoskeleton tracking with model error and human interaction. **Contribution:** GP-residual robust MPC, human-oriented analytical and learned inverse kinematics, force-aware interaction, and mirror rehabilitation. **Hardware:** physical 7-DoF ETS-MARSE. **Stack:** LabVIEW RT/FPGA, C++, GP/MPC. **Result:** 387 microseconds for a 10-step MPC horizon, 5 microseconds for analytical IK, and 43 microseconds for learned IK. Video.

Symbolic Robot Dynamics and Real-Time Optimization

Problem: make reusable robot-dynamics expressions fast enough for real-time control. **Contribution:** built custom RNEA tools for Standard and Modified DH models, branched robots, and automatic common-term grouping. **Result:** approximately 90 percent smaller expressions and 210 microseconds for feedforward torque.

Additional Engineering Work

Cross-simulator tactile mapping through a common ROS 2 interface in Gazebo and MuJoCo, plus a C++/Qt/URDF/Ruckig trajectory application with inverse-kinematics tests up to 1 kHz. See the portfolio for demonstrations and validation context.

TECHNICAL SKILLS

Robotics Foundations: Kinematics and dynamics, inverse kinematics, Jacobians, redundancy resolution, RNEA, trajectory generation

Manipulation and Control: Cartesian, whole-body, torque, admittance, model predictive, robust, and sliding-mode control, plus numerical optimization

Robotics Software and Simulation: C, C++, Python, MATLAB, ROS 2, Pinocchio, URDF, Qt, Ruckig, Gazebo, MuJoCo, Isaac Sim

Real-Time Hardware and Sensing: LabVIEW RT/FPGA, VHDL, BLDC actuation, Hall feedback, force/torque, IMU, EMG, depth sensing, DAQ, digital signal controllers

Robot Learning: Gaussian Processes, learning-based MPC, neural-network inverse kinematics, PyTorch reinforcement-learning experiments in simulation

EDUCATION

2020-2023 | PhD in Robotics: Learning-Based Upper-Limb Robotic Rehabilitation, ÉTS, Montréal

2014-2016 | M.Sc. in Robotics: Hardware-in-the-Loop Simulation of a Robotic Manipulator, Universidad Tecnológica de la Mixteca

2009-2014 | B.Eng. in Mechatronics: Universidad Tecnológica de la Mixteca, CENEVAL EGEL Outstanding distinction

Recognition: 2024 ÉTS Excellence Award nomination, 2020 doctoral scholarship, 2014 master’s scholarship

SELECTED PUBLICATIONS

Bedolla-Martínez, D., Kali, Y., Saad, M., Ochoa-Luna, C., and Rahman, M. H. (2023). *Learning human inverse kinematics solutions for redundant robotic upper-limb rehabilitation.* Engineering Applications of Artificial Intelligence, 126, 106966. DOI: 10.1016/j.engappai.2023.106966.

Bedolla-Martínez, D., Kali, Y., Saad, M., Ochoa-Luna, C., and Rahman, M. H. (2023). *Robust MPC with Integral Super-Twisting for Trajectory Tracking of an Exoskeleton Robot Arm.* IEEE PEDS. DOI: 10.1109/PEDS57185.2023.10246735.